CSC336 · WEB TECHNOLOGIES

Assignment 4
Real Backend

Express REST API, MongoDB database, bcrypt password hashing, JWT authentication, protected order creation, and a frontend connected through fetch.

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01 · REST API

Products from MongoDB

The frontend no longer reads a local JSON file. It requests products from Express, and the server reads the MongoDB **products** collection through Mongoose.

```
GET /api/products
GET /api/products/:id

Response: 200 OK
15 products returned from tesla_store.products
```

Express routes | Mongoose model | HTTP 200 / 404

02 · AUTHENTICATION

Accounts and protected orders

Registration hashes the password with bcrypt. Login compares the hash and returns a JWT. The custom **requireAuth** middleware checks the Bearer token before an order can be created.

```
POST /api/auth/register → 201 Created
POST /api/auth/login → 200 OK
POST /api/orders → 401 without token
POST /api/orders → 201 with valid token
```

Passwords never stored plain | JWT token | User-linked orders

03 · DATABASE SETUP

MongoDB Compass workflow

Start MongoDB, run the seed script once, and open MongoDB Compass to inspect the database. The application creates the collections through Mongoose when data is inserted.

```
Database: tesla_store
Collections: products, users, orders
Connection: mongodb://127.0.0.1:27017/tesla_store

npm install
node seed.js
npm start
```

15 seeded products | MongoDB Compass ready | Frontend served by Express